



# Edge-Based CO<sub>2</sub> Beacon: On-Device Transportation Mode Classification for Real-Time Emission Estimation

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Electrical and Computer Engineering Capstone Design (ECE499)

## Background

- Transportation accounts for 29% of U.S. greenhouse gas emissions (EPA, 2023).
- Existing carbon trackers rely on GPS and cloud processing, raising privacy and battery concerns.
- Goal: Provide real-time, private emission feedback without GPS or cloud dependency.**
- Solution: An embedded beacon that classifies 5 transportation modes (walk, car, bus, train, subway) and estimates CO<sub>2</sub> on-device.*

## Design Objectives & Specifications

- Classification Accuracy:** Overall accuracy of 90% or higher across 5 transportation modes using only inertial and barometric sensor data.
- Inference latency:** On-device classification within 1 second per 8-second window.
- Battery runtime:** 24+ hours of continuous inference and sensing on 1200 mAh Li-Ion.
- CO<sub>2</sub> Estimation Error:** Less than 20% of ground-truth CO<sub>2</sub> per trip, validated over 5 or more trips.
- Privacy:** No raw sensor or location data transmitted via BLE.
- Model Compression:** Model must fit within 796 KB flash on nRF52840 microcontroller.

## Design Description

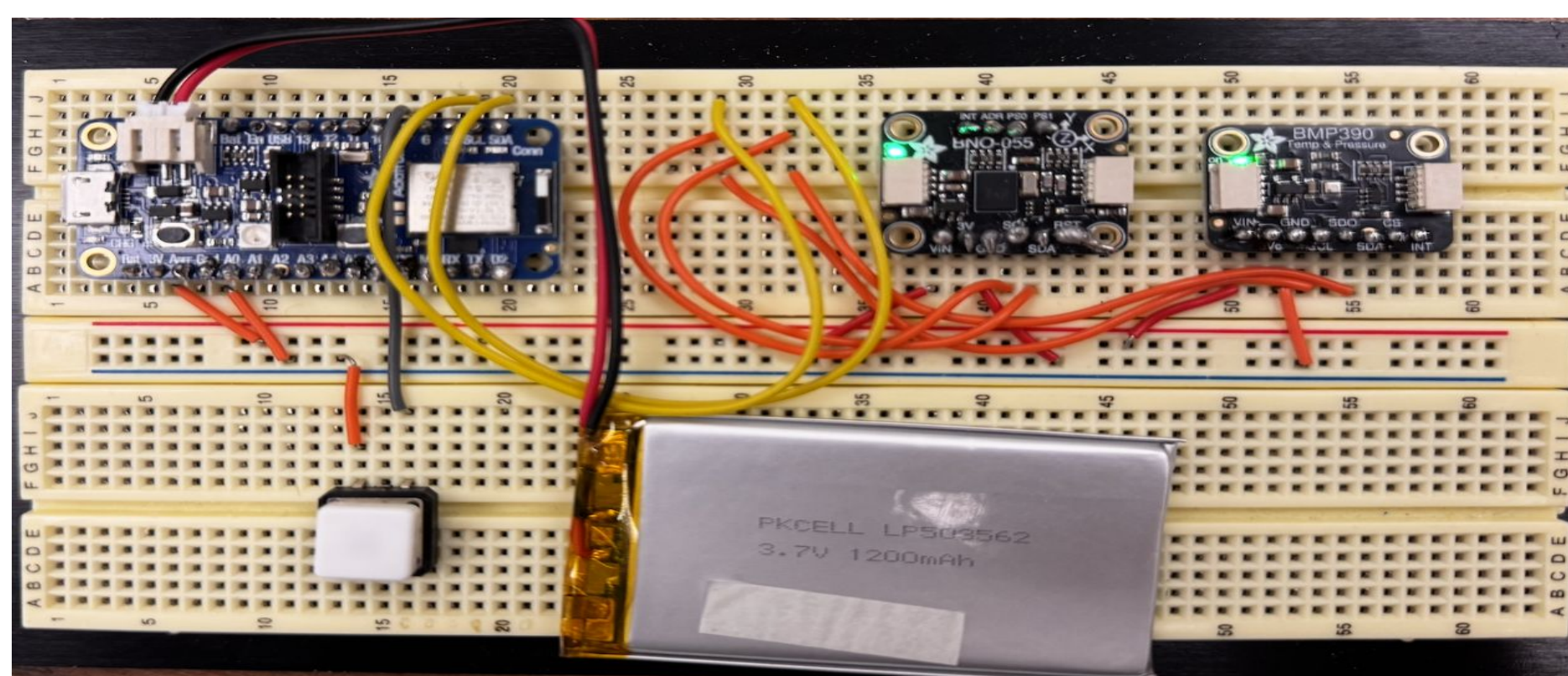


Figure 1: Assembled CO<sub>2</sub> Beacon Prototype

## Design Description (cont.)

- Hardware: Feather nRF52840 Microcontroller, BNO055 IMU, BMP390 barometer, 1200 mAh Li-Ion battery)
- Push button used to start and stop trip segments during data collection.

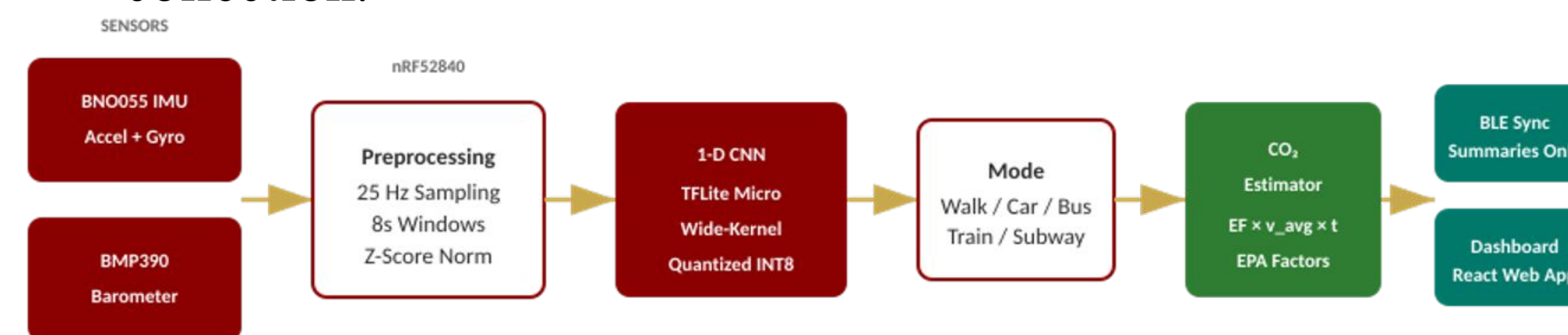


Figure 2: System Overview

- Sensor data (7 channels, 25 Hz) is windowed into 8-second segments, z-score normalized, and fed into an INT8 quantized 1-D CNN running on TFLite Micro.
- Mode classification drives real-time distance and CO<sub>2</sub> accumulation. Trip summaries are transmitted via BLE.

## Validation Results

| Specification                    | Target                                      | Result                  | Status  |
|----------------------------------|---------------------------------------------|-------------------------|---------|
| CO <sub>2</sub> Estimation Error | ≤ 20% of ground-truth CO <sub>2</sub> /trip | Mean 34.0%<br>(5 trips) | Not Met |
| Classification Accuracy          | ≥ 90%                                       | 92.7%                   | Met     |
| Inference Latency                | ≤ 1 second                                  | 261.6 ms                | Met     |
| Battery Runtime                  | ≥ 24 hours                                  | 48+ Hours               | Met     |
| Privacy Preservation             | No raw data over BLE                        | By architecture         | Met     |
| Model Compression                | Fit within 796 KB flash                     | 361 KB                  | Met     |

Self-only CNN (v4 wide-kernel) Confusion Matrix (Acc=0.927)

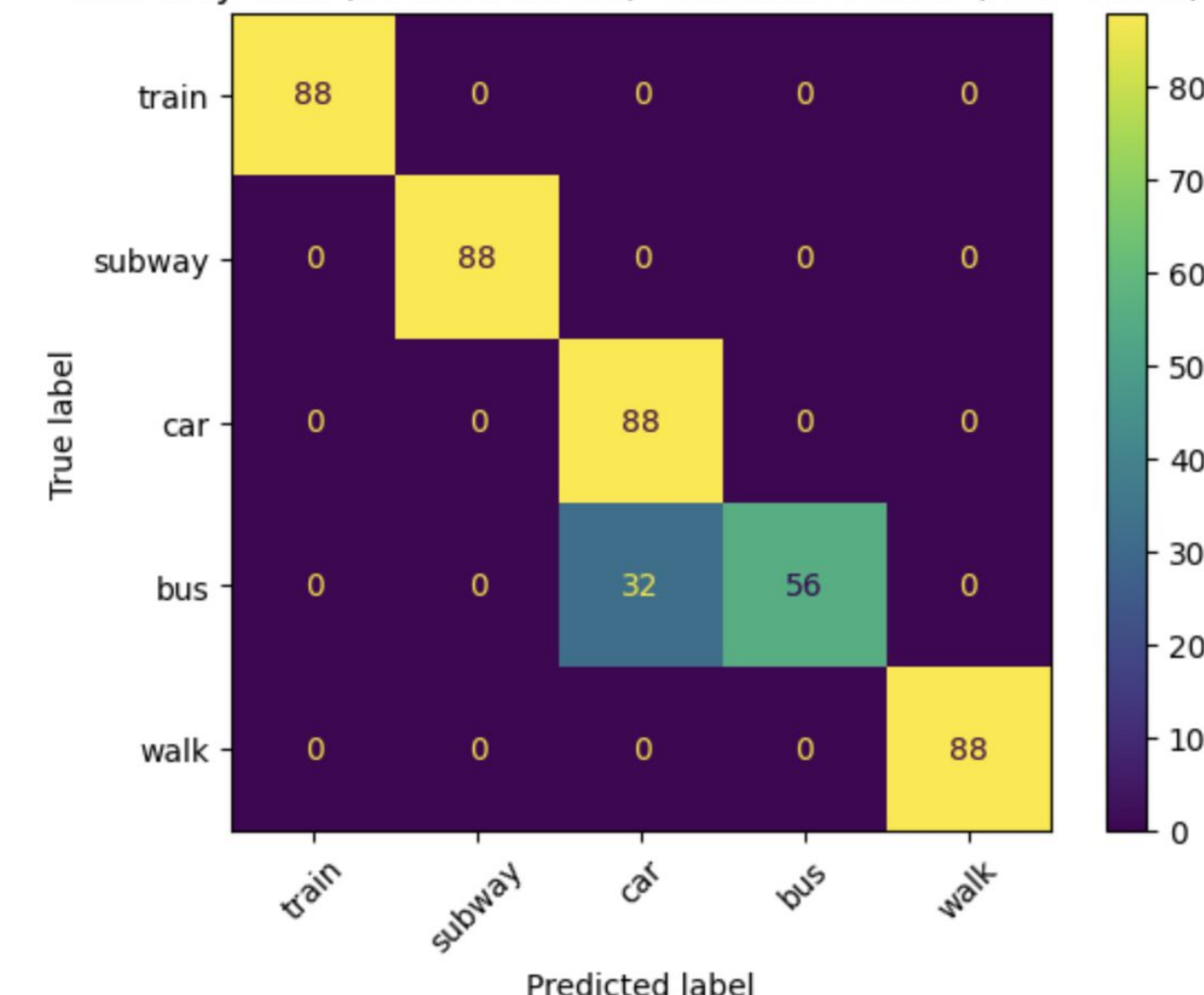


Figure 3: Per-class Confusion Matrix for 5 Modes of Travel

Model Compression: Float32 → INT8

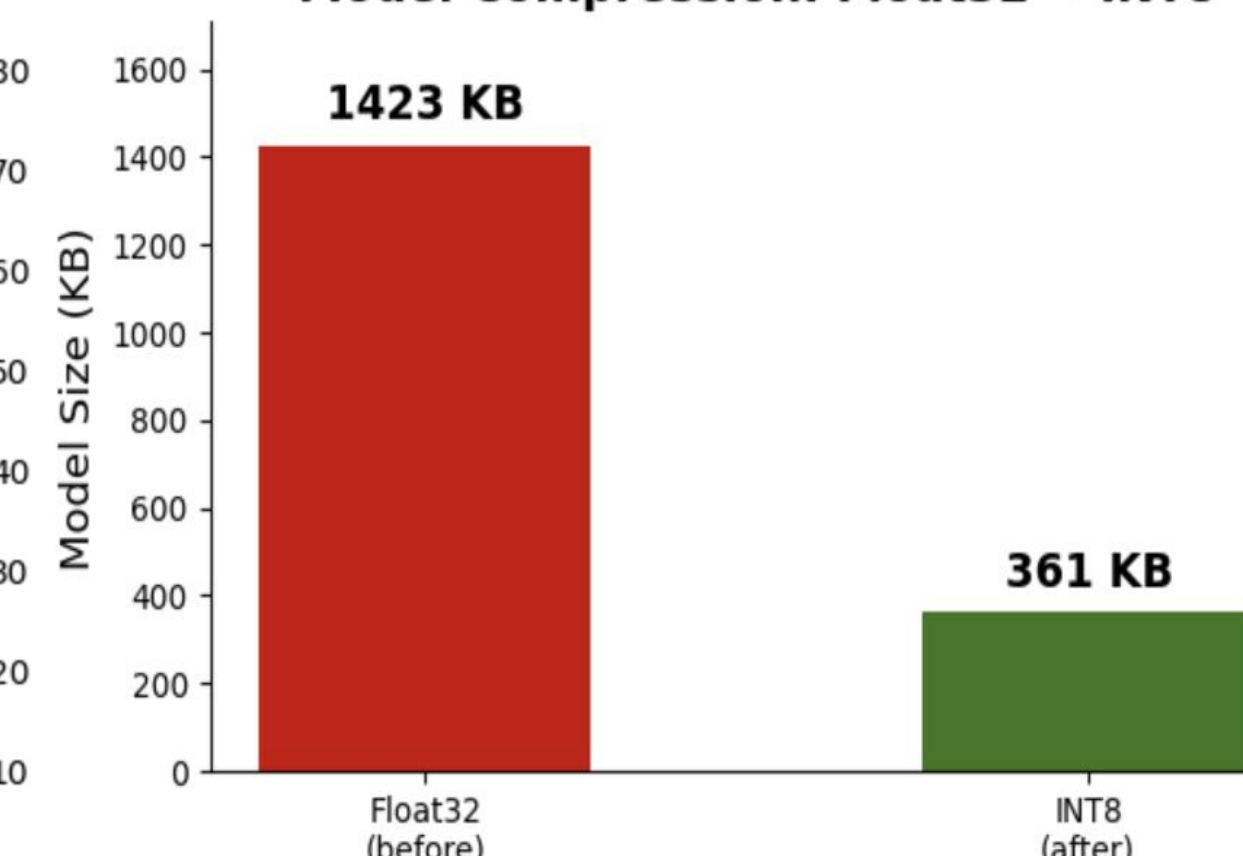


Figure 4: Model Compression Bar Chart

## Results (cont.)

```
--- Latency Summary (n=30) ---
Mean: 261.6 ms
Stdev: 0.7 ms
Min: 261 ms
Max: 264 ms
Median: 262.0 ms
```

Figure 5: Latency Summary

- Classification Accuracy:** 92.7% overall across 5 modes.
- Inference Latency:** 261.6 ms mean (n=30, σ=0.7 ms).
- Battery Runtime:** 48+ hours continuous on 1200 mAh Li-Ion.
- Privacy:** No raw sensor or location data transmitted over BLE.
- Model Compression:** 361 KB INT8 quantized model fits within 796 KB flash.
- CO<sub>2</sub> Estimation Error:** Mean 34.0% over 5 car trips. Fixed-speed assumption (40 km/h) is the dominant error source.

## Discussion & Conclusions

- The CO<sub>2</sub> beacon demonstrates that 5-mode transportation classification is achievable entirely on a low-power embedded device with no GPS or cloud dependency at 92.7% accuracy and 261.6 ms latency..
- Bus/car confusion is benign: both modes share identical EPA emission factors (0.089 kg/km), producing near-zero estimation error from misclassification.
- The fixed-speed CO<sub>2</sub> pipeline is the system's primary limitation. Actual trip speeds (25-54 km/h) diverged from the 40 km/h assumption, yielding 34.0% mean estimation error across 5 trips.
- Future work: Add more transportation modes, miniaturize into wearable enclosure, GPS-augmented distance sensing to resolve CO<sub>2</sub> accuracy.

## Acknowledgements

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